Divyam Sengar

San Diego, CA
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SKILLS

Java, Python, C, C++, Go, R, SQL, MATLAB, HTML/CSS, JavaScript, CUDA, Jetbrains, Git, JUNIT, PlayCrypt, React, Express, MongoDB, Node, Android, Roboelectric, RoomDatabase, Docker, GCP, Kubernetes, PyTorch, sklearn, TensorFlow, Pandas, Valgrind, GDB, Nachos, Requests, gRPC, Convex optimization, Transformer, TabTransformer, AWS, Numpy, Scipy, OpenSearch, Agentic AI, IaC, RAG, Secure AI, Lattice Algorithms, Parallelization & Vectorization, Quantum Algorithms

WORK EXPERIENCE

Computer Science and Engineering Society @ UCSD — **Jun 2023 - Sep 2023**
Full Stack Developer Intern — *San Diego, CA*

- Developed a dynamic membership platform website using React.js and Node.js, enabling secure user authentication and increasing membership engagement by 50%
- Implemented the backend with Express.js and MongoDB to efficiently store and manage user & event data

UC San Diego Computer Science and Engineering — **Sep 2023 - Present**
Instructional Assistant — *San Diego, CA*

- Mentored 500+ student classes for Python Programming, Advanced Data Structures, Computer Organization, and Deep Learning
- Conducted tutoring over 20+ weekly office hours, clarifying concepts and bolstering student coding proficiency
- Produced homework, exams, and dynamic question generators, improving assessment of student comprehension

DataAnnotation — **Jun 2024 - Sep 2024**
Software Engineer Intern — *Remote*

- Optimized AI chatbot performance by designing and implementing fine-tuning strategies for transformer-based models, resulting in improved code correctness and efficiency.
- Engineered and validated training pipelines for large language models, focusing on the generation of diverse and challenging coding problems across multiple languages to enhance model reasoning.
- Conducted in-depth analysis of model architecture, identifying key bottlenecks and implementing data-driven optimization techniques to reduce latency and improve output quality.
- Evaluated the performance of various chatbot models with respect to differential coding tasks, focusing on key performance metrics such as code correctness, brevity, and execution time.

Amazon Web Services — **Jun 2025 - Sep 2025**
Software Development Engineer Intern — *Bellevue, WA*

- Fully Architected and Developed OSCAR, an open-source serverless multi-agent AI Chatbot for automating OpenSearch release management and lowering open-source community barrier to entry. The project was highlighted by the Linux Foundation @ Open Source Summit, presented @ the OpenSearch Conference, and earned formal recognition from multiple AWS VPs as an "AWS AI win".
- Engineered a sophisticated multi-agent framework using AWS Bedrock, establishing agent-to-agent (a2a) methods to facilitate seamless communication between a supervisor agent and specialized agents dedicated to discrete workflow automation tasks.
- Implemented a novel dual-agent security architecture using RBAC, providing secure, authorized natural language Jenkins job executions, while keeping authorization invisible for a frictionless experience.
- Engineered a comprehensive feature set, empowering users to trigger Jenkins jobs with parameter validation, retrieve esoteric release knowledge, query OpenSearch clusters for real-time test, build, and release metrics, and automate targeted stakeholder communications & pinging.
- Designed and deployed Infrastructure as Code using AWS CDK with modular stack architecture, managing Bedrock Agents, DynamoDB tables, Lambda functions, Knowledge Base Resources, API Gateway, Secrets Manager, and IAM roles with least-privilege security principles.
- Architected a bespoke RAG system, using a custom OpenSearch VectorDB to retrieve context-specific knowledge from data streams and knowledge docs in an s3 bucket.
- Created a highly scalable, serverless, event-driven architecture designed for resilience and performance. The system used an API Gateway to handle and queue Slack events asynchronously, preventing throttling, and leveraged auto-scaling AWS Lambda functions with comprehensive CloudWatch monitoring.
- Enhanced user engagement by implementing robust, multi-turn conversation context management using DynamoDB with TTL-based session storage. This was complemented by a rich, cross-channel communication system in Slack, featuring slash commands, interactive UI components, and real-time status updates via emoji reactions.

UC San Diego Computer Science and Engineering — **Jan 2026 - Mar 2026**
Teaching Assistant — *San Diego, CA*

- Only TA for a class of Discrete & Continuous Optimization, harboring 100+ students

PROJECTS

Data Compressor | C++

- Developed a custom Huffman file compression & decompression tool in C++ for efficient, lossless data compression
- Minimized compression overhead and optimized compression speed by utilizing custom data structures
- Reduced file size by 60% on average, applying a frequency based encoding approach

QuoteGuard | Python, PyTorch

- Developed a custom transformer-classifier to verify speakers of political quotes & speeches
- Achieved 96.2% accuracy by leveraging custom sparse attention mechanisms and positional embeddings
- Optimized accuracy via regularization, cross-validation, and varying the feed forward layer normalization functions

Diabetes Prediction | Python, sklearn, Pandas

- Codeveloped a diabetes prediction model using physiological and demographic data, achieving 96.3% accuracy
- Implemented and optimized SVM, XGBoost, Logistic Regression, and Random Forest models
- Optimized model performance via hyperparameter tuning and cross-validation techniques to reduce overfitting
- Decimated false positives via data augmentation to engineer cross-validation datasets
- Developed a new TabTransformer architecture to improve accuracy on categorical features, improving overall F1 score

Successorator | Java, SQL, Android, Git, Github Actions, XML

- Designed and implemented a multifaceted to-do list Android app using Java, with recurrent tasks, focus mode, categories, calendar synchronization, persistence, deletion, rollover, and task preemption
- Used SQL & RoomDatabase to implement persistent features, speedy data retrieval, and enable task rollover logic
- Customized MVP architecture to streamline app workflows, ensuring intuitive and fast user input
- Built a robust Github Actions test suite, ensuring proper merge protection and automated testing

Welp | Golang, Python, C, Docker, Kubernetes, GCP, gRPC

- Built a Yelp-like microservices platform with scalable Detail, Review, and Reservation services using Golang and gRPC, deployed via Docker/Kubernetes with automated CI/CD pipelines
- Designed latency-throughput analysis, identifying frontend bottlenecks and improving max throughput by 4.2x via Kubernetes pod scaling.
- Integrated LRU/LFU caching policies to reduce storage-layer access latency by 85% under Zipfian workloads, achieving sub-10ms cache-hit response times
- Engineered a C-based thread scheduling and affinity benchmark to analyze concurrency primitives, manipulate thread-to-core affinity (via Linux system calls), and optimize cache locality, thus reducing cross-core memory latency by 10.8ms on GCP machine and improving performance through same-core co-location.

Privacy Impact Indicator & AI Platform Privacy Analysis | Chrome MV3, JavaScript, Python, Selenium, Scrapy, Pandas

- Designed and developed Privacy Impact Indicator, a Chrome Manifest V3 extension that visualizes real-time web tracking risk using an exponential privacy scoring model
- Engineered a browser-side telemetry system using service workers, ring-buffered request logs, and peripheral UI feedback, nudging users away from tracker-heavy sites without blocking traffic
- Conducted a 15-participant within-subjects user study, reducing tracker contact by 32% and third-party requests by 14% with no meaningful increase in NASA-TLX workload
- Built a Selenium–Scrapy web crawler to audit cookie and local-storage practices on major AI platforms (ChatGPT, DeepSeek, X, Facebook)
- Analyzed 157 cookies across platforms, revealing widespread use of long-lived first-party cookies (avg. 293 days) and incomplete adoption of SameSite, Secure, and HttpOnly flags
- Validated privacy risk metrics against Privacy Badger grades (AUROC 0.91, Spearman Á= 0.81), demonstrating strong alignment with independent tracker assessments

EDUCATION

University of California, San Diego **Sep 2022 - Mar 2025**
Bachelor of Science in Computer Science

University of California, San Diego **Jan 2025 - Mar 2026**
Master of Science in Computer Science

Total Relevant Coursework:
Advanced Data Structures, Algorithm Design & Analysis, Computability & Complexity, Computer Architecture, Computer Networks, Computer Security, Convex Optimization Formulations and Algorithms, Data Center Systems, Deep Learning, Digital Systems, Discrete & Continuous Optimization, Human-AI Interaction, Modern Cryptography, Numerical Analysis: Approximation & Nonlinear Equations, Numerical Linear Algebra, Object Oriented Development, Operating Systems, Parallel Computing, Programming Languages, Quantum Complexity & Cryptography, Quantum Computing, Recommender Systems & Web Mining, Software Engineering, Statistical NLP, Theory of Computation, Machine Learning with Music, Probabilistic AI, CyberSecurity & Privacy